

5ARF0 Assignment Description

In one sentence: You will make a design strategy and a demo/ mock-up for an AI-based tool with an educational and/or ethical function.

In brief: In groups of four, students create a design strategy and a demo/mock-up demonstration for an AI-based tool on one of the following three topics:

- AI for Animal Welfare [[ANWE](#)]. (Design an AI-driven animal-welfare management tool for a chosen context (e.g., agriculture, wildlife, or lab research) that safeguards animals' interests while avoiding species bias, objectification, and over-datafication.)
- AI for Eating Disorder Symptoms [[EADS](#)]. (Design and prototype an AI-integrated interface for an automated eating-disorder support system like Featback (featback.nl), while critically examining the ethical risks (user harm, misuse, bias) it raises.)
- AI for Environmental Sustainability [[ENSU](#)]. (Draft an AI-powered Sustainability Tool for TU/e (in Research, Education, or Operations) that creatively extends the existing self-assessment tool, aligns with TU/e's Sustainability Vision, and applies AI to advance sustainability.)

On CANVAS, you will find a page on each of these topics with resources and additional information. However, you should not limit yourself to these resources.

The design strategy explicitly takes into account the following elements:

- Embodiment or embedding in a specific educational or ethics context
- Formatting and content of the outputs of the tool, and satisfaction of relevant benchmarks for output quality
- Ability to tweak or modify the tool with the participation of the subject expert(s)
- Perceived value of the generated outputs for students, clinicians, researchers, or animal advocates.

The demo and/or mock-up will allow domain experts to assess generated outputs, and to experiment with the setup and give input to the design. You may add elements to the demo/ mock-up such as study questions, links, a chatbot, an embedding (e.g., on a webpage or in an avatar), etc., if they fit with the scope of the assignment and contribute to the above elements.

The written assignment will be assessed primarily on awareness of ethical concerns raised by different approaches, motivation for iterative solutions to the problem, and sophistication in identification and anticipation of limitations and barriers in the target context. Clarity and compellingness of the design solution will also be assessed.

Students will make a presentation and will write two reports and a short reflection and peer evaluation. Every 2-3 weeks there will be a meeting with coaches, usually together with one other group. At the end of the term, you practice and prepare a video clip of a short pitch of your design concept.

Set-up

In order to solve the problem, you will need to apply your creativity and design skills whilst taking into consideration a target context and domain expert perspectives. Your answer to the challenge can take many forms. You will communicate your response to domain experts via a report, a demo/ mock-up, and a video clip of a pitch presentation.

The report will clearly state the problem you identified and the proposed response. It will also detail the identified functional and non-functional requirements and the selection and application of suitable methods to solve the problem and the presentation of your results. This is detailed in the table below.

Overall Process	Design Process	Ethics Process
1-Identify the problem you will solve.	Identify the core design problem and its scope.	Define the problem by looking at the stakeholders, their interests and their values.
2-Identify functional and non-functional requirements.	Identify functional and non-functional requirements for the design problem.	Identify functional and non-functional requirements domain experts and stakeholders. What do experts expect?
3-Select and apply a suitable method to solve the problem	Select and apply suitable concepts and frameworks, methods and tools for solving the defined problem.	Select and apply suitable concepts and frameworks, methods and tools to identify, balance, and promote values in the target context, resolving any conflicts in a transparent way.
4-Present results suitable for the audience	Present results and motivation at a level that non-specialists can understand, while communicating technical options.	Present results that meet the informational needs of the audience, responding to their ethical concerns.
5-Demonstrate self-reflective awareness of influences between design and ethics	Demonstrate self-reflective awareness as a designer of the technical and ethics constraints on AI design problems.	Demonstrate self-reflective ethical awareness of influences between AI design and ethics.

Although the above table implies a linear flow to solving the challenge you will iterate overall, or some, of these stages more than once.

To help keep your team on track towards a final solution and to give regular feedback on your progress you will be supported by the instructor.

Learning Support: How will you be supported?

You will receive support for your work during 4 types of weekly contact moments. All these moments are ON CAMPUS.

Lectures 4h per week	Each week you will attend 4 hours of ethics lectures and supporting in-class activities. You will practice applying ethics to your case and the work you will do here will be directly relevant for your assignment.
Peer review, coaching and instructor feedback 15 min/ week	Your team will give and receive feedback from other teams regarding their, and your, approach to solving challenges. The instructor and coach will facilitate this in coaching meetings and will also give feedback. There will also be a peer review opportunity on written material. (See the course schedule.)
Tutorial and work time 2h per week	Your team will gather to work on the challenge. During this session, you will be able to discuss questions with instructors and gain insight into your progress.
Presentation sessions	In weeks 7-8, you will present and receive feedback from other groups and from the instructor.

PLEASE SEE CANVAS MODULES FOR A WEEK-TO-WEEK SCHEDULE

Assessment and Grading

Assignment Phase 1: At the end of Week 3, you will turn in an initial report describing the motivation and goals of the project, identifying requirements, and anticipating issues that need to be addressed. See the suggested format document at the Assignment Page on Canvas.

Inter-group Peer Review: In weeks 5-6, you submit a draft version and give feedback to another group on their draft version.

Presentation: In weeks 7-8, you will present the results of your project so far to the class (15 minutes).

Assignment Phase 2: At the end of week 8, you will turn in a final report. This will provide a more detailed rationale for your strategy and the choices you made during the course, and a detailed discussion of ethical issues raised in the project. See the suggested format document at the Assignment Page on Canvas.

Groupmate Evaluation: At the end of week 8, you will give an evaluation of your groupmates and write a short reflection. Adjustment of individual scores will be based on the groupmate evaluation.

Assignment Phase 1	10%
Inter-group peer Review	---
Presentation	10%
Assignment Phase 2	70%
Groupmate Evaluation	10%
Total	100%

Resit

Only the in-class exam is subject to the possibility of a resit in this course.

Support and Mitigating Circumstances

The structure and assessment of this course requires cooperation and communication between all group members. As an individual, it is your responsibility to keep your team notified if your circumstances prevent your full participation. As a team, it is your responsibility to communicate with all team members equally and ensure all individuals play an equal role.

Resolution of internal cooperation and coordination issues should proceed as follows:

1. In the first case by internal group communication. You should make at least 2 attempts to communicate your concerns and give some time for a response (this will be dependent on the gravity of the situation).
2. If there is a failure to resolve the issue internally, then you should contact your instructor including relevant screenshots/emails of attempts to communicate. This should be done by email.
3. Alert the coaches and/or instructor as required.

Advisory: Keep a note of the date/time and names of those in attendance at all group meetings. Any change in personal circumstances (mitigating circumstances) that has, or may have, an immediate or future effect on your engagement and participation in the course **MUST** be reported to the instructor. Once your work has been submitted (both formatively and summatively) you will not be allowed to apply retroactively for circumstances to be taken into account.

For personal support (i.e. illness, bereavement) you should also contact your [Academic Advisor](#). You can also use the [Education and Student Affairs](#) flowchart to determine who to contact.

Academic Integrity

Ground rules for students

Be aware that you should always operate with academic integrity, taking into account the rules and regulations about copyright, plagiarism and fraud. Use the links below to learn more about what you are (not) allowed to do in your assignments and assessments.

TU/e Rules for AI use in Education	EE BSc Regulations and PER
TU/e Regulations and Code of Conduct	EE MSc Regulations and PER
TU/e Copyright Information Point	AI&ES MSc Regulations and PER

On the use of AI

Any work for an assignment in which generative AI tools are allowed, should make unambiguously clear:

- which tools you used,
- for which parts of the work you used them,
- what you used them for.

You should be critical about the quality of your information sources and make sure to precisely reference the parts of your work where AI tools were used.

In any use of AI, you remain responsible for the work you deliver, down to the individual sentence.

Use of AI other than that indicated may lead to a suspicion of fraud.

AI is the object of your assignment, not the process by which you do the assignment. AI is not a wise means of explaining your process of working with that object or justifying your thought

process. You may use it in superficial ways to brainstorm ideas or do a spelling and editing check, at your own risk. You should maintain complete critical control over its use. For this reason, you are not allowed to use generative AI to compose the text of your assignment. You are allowed to use AI tools as a source for inspiration or to improve/enhance the quality of products, although this is not recommended. If you do choose to use these tools, you should be able to explain why you opted to use specific tools and evaluate their usefulness.